MILITARY SPECIFICATION

ALTIMETER-ENCODER, AAU-21/A

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 This specification covers the requirement for one type of counter-drum-pointer integrally lighted self-contained altimeter-encoder (indicator) having a range of -1,000 to +38,000 feet and a 115V AC, 400 hertz servo driven encoder.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors and other uses

Military

MIL-P-116	Preservation, Methods of
MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum Alloys and Aluminum
MIL-D-1000/1	Drawings, Engineering and Associated Data
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for

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MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-B-27497	Bearing, Jewel, Sapphire, or Ruby Synthetic

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-470	Maintainability Program Requirements
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics, and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements for reliability program
MIL-STD-794	Parts & Equipment, Procedures for Packaging and Packing of
MS 3113	Connectors, Receptacle, Electric, Solder Mounting, Miniature
MS 33549	Case, Aircraft Instrument, 2 3/4 inch dial with sump, Standard Dimensions for
MS 33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS 33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS 33649	Bosses, Fluid Connection, Internal Straight Thread

PUBLICATION

Air Force

U.S. Standard Atmosphere 1962

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Federal Aviation Agency

U.S. National Standard for Common System Component Characteristics for the IFF Mark X (SIF)/Air Traffic Control Radar Beacon System, SIF/ATCRBS

Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120

(2) Copies of the FAA document may be obtained from the Federal Aviation Agency, Washington, D.C. 20553

3. REQUIREMENTS

3.1 Qualification - The instruments furnished under this specification shall be a product which has been tested and meets the quality assurance provisions specified herein and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of Specifications and Standards - Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

3.3 Standard General Requirements

3.3.1 Non-magnetic Materials - Non-magnetic materials shall be used for all parts except where magnetic materials are essential. The extent to which magnetic materials are required shall be reported to the procuring activity.

3.3.2 Protective Treatment - When materials are used in the construction of the altimeter-encoder that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service life, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of the specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.3 Toxic and Corrosive Fumes - The materials, as installed in the altimeter-encoder and under the conditions specified herein, shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

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3.3.4 Aluminum Alloy Parts - Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that the dial, small holes, and case inserts, need not be anodized. Parts which do not anodize satisfactorily may be coated in accordance with MIL-C-5541.

3.3.5 Components - The electronic and mechanical components shall be in accordance with MIL-E-5400.

3.3.6 Jewel Bearings - All sapphire and ruby jewel bearings used in the altimeter shall be in accordance with MIL-B-27497.

3.3.7 Threads - The threads of all machine screws shall conform to MIL-S-7742.

3.4 Design and Construction

3.4.1 The altimeter-encoder shall include the following features: a precision all mechanical altimeter mechanism, a counter-drum-pointer altitude display, integral red & white lighting, a barometric setting displayed by a four digit veeder type counter, and a servo driven encoder whose position corresponds to the altitude displayed when the baro setting is at 29.92. The altimeter-encoder shall be so constructed that no parts will work loose in service and will withstand the normal shocks, vibrations, and such other conditions as are incident to service, shipping, storage, and installation without failure.

3.4.2 Maintainability - The contractor's maintainability program shall be in accordance with MIL-STD-470. The maintenance concept for this equipment shall be repair at depot level only.

3.4.2.1 Repair Time - The altimeter-encoder shall be completely repairable at depot level. The maintenance time shall not exceed 8 hours utilizing one depot technician and authorized depot AGE and technical information. Spare parts and facilities are assumed to be available. Maintenance time shall include all repair and test time excluding government acceptance testing.

3.4.3 Mechanism Adjustment - The mechanism shall be provided with a means of adjusting or correcting the indications to conform to the performance required herein, if this should become necessary during the life of the altimeter. The calibration adjustment shall not affect the temperature performance of the instrument. This means of adjustment shall be reasonably simple to manipulate and of such nature that adjustment can be accomplished by tools ordinarily used by an instrument repairman.

3.4.4 Barometric Setting System - A setting system, controlled by a knob, shall be provided that will permit the altimeter display to be set to indicate zero altitude at any existing ground level pressure throughout the range of 28.10 to 31.00 (inclusive) inches of mercury in 0.01 inch increments. The shaft of the setting knob shall be capable of withstanding a load of 15 pounds applied perpendicular to the shaft

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at the extreme outward end of the knob without permanent deformation. A positive stop or stops shall be provided whereby the barometric zero setting mechanism cannot be moved beyond the range of 28.00 to 31.10 inches of mercury. The stops shall be capable of withstanding a torque of 2 pound inches applied to the barometric setting knob without damage to the stops or instrument mechanism. The resistance when setting the barometric pressure shaft shall be 5±2 ounce inches. The barometric counter unit displaying the setting shall be located as shown in Figure 1. The mechanism shall be capable of withstanding a pull of 15 pounds while the barometric scale is rotated throughout the range of settings from the minimum stop to the maximum stop without disengagement or damage of the mechanism.

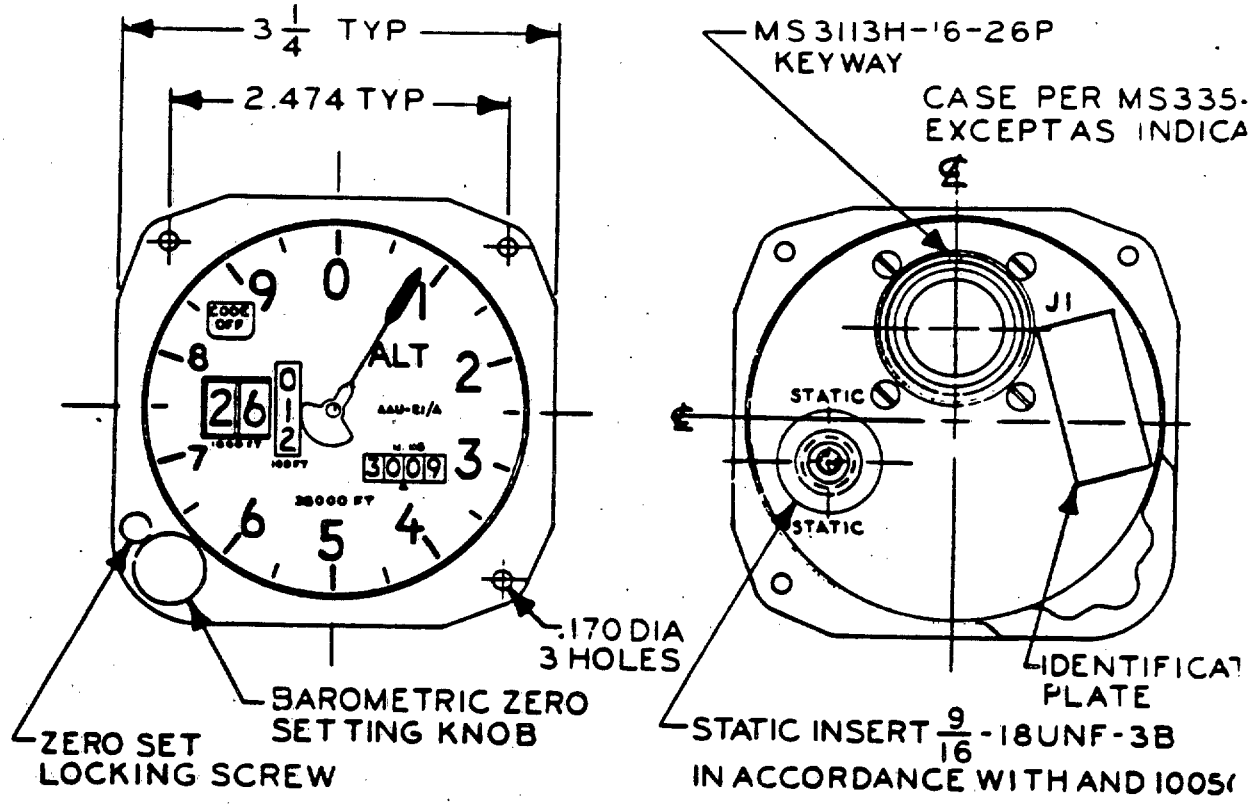
3.4.4.1 Adjustment of Barometric Scale Counter to Pointer - The barometric setting system controlled by the setting knob in its extended position shall permit adjustment of the setting of the pointer to the barometric scale counter.

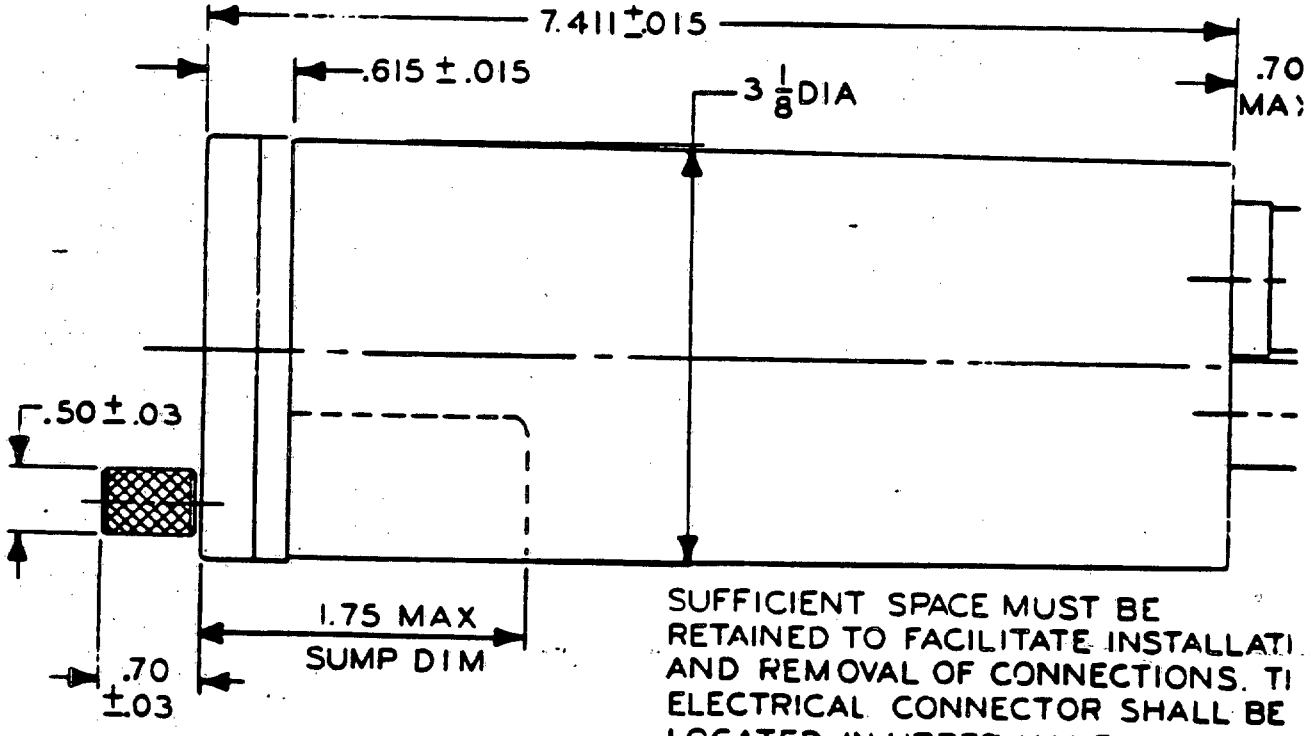
3.4.4.2 Setting knob position and lock - The barometric setting shall be provided with a lock which shall hold the setting knob in its normal position and allow adjustments as specified by para. 3.4.4. The means for unlocking the barometric setting system shall be accessible from the face of the altimeter-encoder when installed on the instrument panel. The unlocked barometric setting system shall allow adjustments as specified in para. 3.4.4.1. The design shall provide for a seal to prevent unauthorized adjustment.

3.4.5 Pointer, Counter, and Drum Relationship - The pointer shall indicate readings in feet with reference to a single circular uniform graduated scale making one revolution for each 1,000 feet of altitude. The pointer shall be positively geared to the 100 ft. drum so that as the pointer moves CW (increasing altitude), the drum will rotate for increasing numbers synchronized with the pointer. The numerals on the 100 ft. drum shall represent hundreds of feet. The 1,000 ft. drum shall be driven by the 100-foot drum so that the 1,000 ft. drum will index one digit as the 100 ft. drum moves through the 9 to 0 position in either direction. During the long period between 0 and 9, the 1,000 ft. drum shall be locked in position and in alignment with 10,000 ft. drum. Each digit on the 1,000 ft. drum shall represent 1,000 ft. intervals. The 10,000 ft. drum shall be driven by the 1,000 ft. drum so that the 10,000 ft. drum will index one digit as the 1,000 ft. drum moves through the short period between 9 and 0 in either direction. During the long period between 0 and 9 of the 1,000 ft. drum, the 10,000 ft. drum shall be locked in position. Each digit of the 10,000 ft. drum shall represent 10,000 foot levels. In the space corresponding to zero, the 10,000 foot drum shall have striped lines, as shown in figure 2. The 1,000 ft. drum and the 10,000 ft. drum are collectively called the counter.

The combined readings of the counter-drum shall indicate the pressure altitude in thousands and hundreds of feet. The pointer shall act as a vernier of the hundreds drum as well as being an indication of

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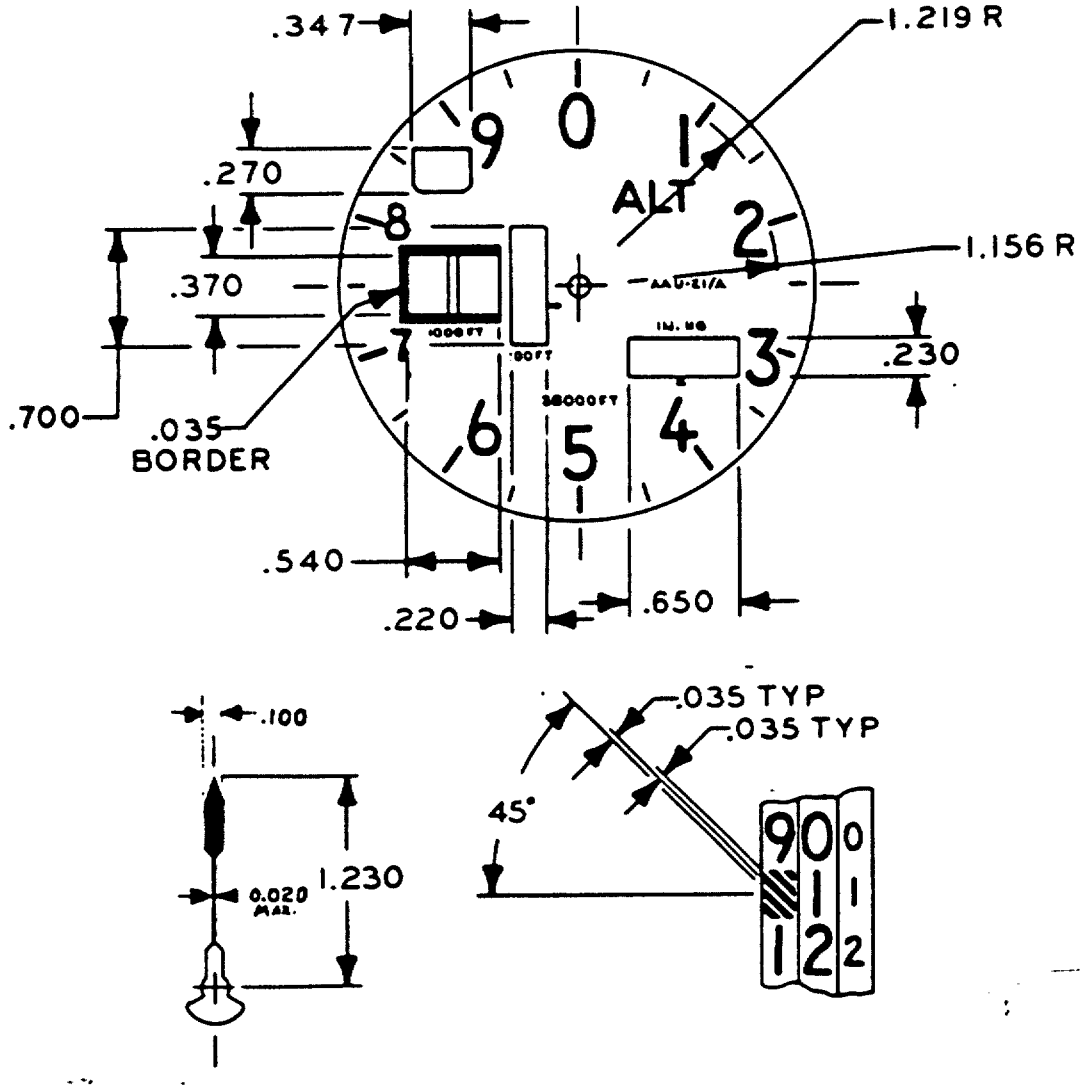


TOLERANCES WHERE NOT SPECIFIED:
DECIMALS ± 0.010
FRACTIONS ± 1/64

SUFFICIENT SPACE MUST BE RETAINED TO FACILITATE INSTALLATI AND REMOVAL OF CONNECTIONS. TI ELECTRICAL CONNECTOR SHALL BE LOCATED IN UPPER HALF AND ON VERTICAL ℄ STATIC CONNECTION SHALL BE LOCATED IN LOWER LEFT QUARTER

FIG. 1

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TOLERANCES WHERE NOT SPECIFIED
DECIMALS ± 0.010
ANGLES ± 1.5°

FIG. 2

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trend information. The counter-weight of the pointer shall not in any position obscure the indication of the drum.

3.4.5.1 Pointer Adjustment - The pointer and drum shall be free to swing to either side of their zero position, without engaging a mechanical stop. The pointer shall be firmly attached to the pointer shaft to prevent slippage under normal service use and still be removable for depot service maintenance.

3.4.6 Case - The outline dimensions and the external connections of the altimeter-encoder case shall conform to Figure 1. The body and mounting flange of the case shall be made of lightweight non-magnetic metal, uniform in texture. The body shall be finished in a durable black finish and the mounting flange shall be finished with a durable lusterless black finish Color No. 37038 of FED-STD-595.

3.4.6.1 Static Connection - The static connection shall be in accordance with Figure 1 and shall be located as shown thereon. A 150-wire mesh filter screen (either stainless steel or monel) shall be firmly installed in the static connection to prevent dirt particles from entering the case. It shall be provided with a vented plug of acceptable material and design. Each plug shall incorporate a vent consisting of a hole 1/16 ± 1/64 inch in diameter.

3.5 Primary Electrical Input Power Requirements - The equipment shall be designed to meet all applicable requirements of MIL-STD-704 and give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as defined therein except that the frequency range shall be 320 to 480 hertz. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V, Category "B", 25 VA

(2) Power 5.0 ± 0.1 V AC or DC 4.0 Watt (Integral Lighting)

(3) 28 V DC ± 1 (Integral Vibrator)

3.5.1 Electrical Connector - The electrical connector used shall conform to Specification MIL-C-26482 and the type specified in Figure 1, and shall be connected as specified in Figure 3. Connector shall be protected with a suitable cap.

3.5.2 Overload Protection - All parts and circuits of the altimeter-encoder which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage. The use of fuses and other protective devices shall not be permitted.

3.5.3 Undervoltage Protection - The altimeter-encoder shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage is within the specified limits.

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3.5.4 Warm-up Time - The maximum warm-up time for the equipment shall not exceed one minute under standard conditions and two minutes at extreme service conditions.

3.5.5 Electrical Outputs - The encoded altitude output shall at all times be referenced to the standard pressure altitude of 29.92 inches of mercury sea level. Operation of the warning flag and the output of the encoder shall not be affected by operation of the barometric setting mechanism. Means shall be provided to re-zero the encoder output to the referenced altitude without removing the mechanism from the case.

3.5.6 Encoder - The digital code shall be in accordance with the U.S. NAT Std for Common System Component Characteristics for SIF/ATCRBS and shall provide a 10-bit 11-lead digital output of pressure altitude. The encoder shall be capable of providing outputs from -1,000 ft. to 38,000 feet in 100-foot increments, with the switching points at altitudes 50 feet ± 40 feet removed from the nominal indicated transmitted altitude. The encoder contact shall be capable of carrying 3.5 ma steady state conditions and an additional 1.5 ma under transient conditions. They shall also be capable of operating under a max. potential of 30V DC. The digitizer, energized by the aircraft transponder, shall be compatible with the Mode "C" input requirements of the transponders.

3.5.6.1 Diode Isolation - The encoder shall be provided with diode isolation. The isolation shall be so wired into the encoder as to block the flow of current if the + side of a d-c voltage source is connected to the encoder common lead and the negative side of d-c connected to any other lead.

3.5.6.2 Forward Voltage Drop and Contact Resistance - The maximum forward voltage drop between each encoder output lead and common shall not exceed 1.5 volts plus 1 volt for each ma of current flow. The total contact resistance between each encoder output lead and common shall not exceed the following values:

Temp (°C)	Effective Resistance (Ohms)	Current (MA)
Room	1500	2.5
Room	2000	1
-55	1700	2.5
-55	2300	1

The minimum resistance for each open circuit shall be 100,000 ohms.

3.5.7 Monitor - A relay shall be incorporated that, in the event of loss of 115 volts, 400 cps power, the common lead to the digitizer shall open. An auxiliary set of single pole double throw contacts (see Figure 3) shall be provided for monitoring power failure. Provisions shall also be made to simultaneously operate the warning flag in the altimeter-encoder when a failure is sensed. Normal power fluctuations as defined in MIL-STD-704 shall not cause nuisance trips. The maximum time delay shall not exceed 3 seconds.

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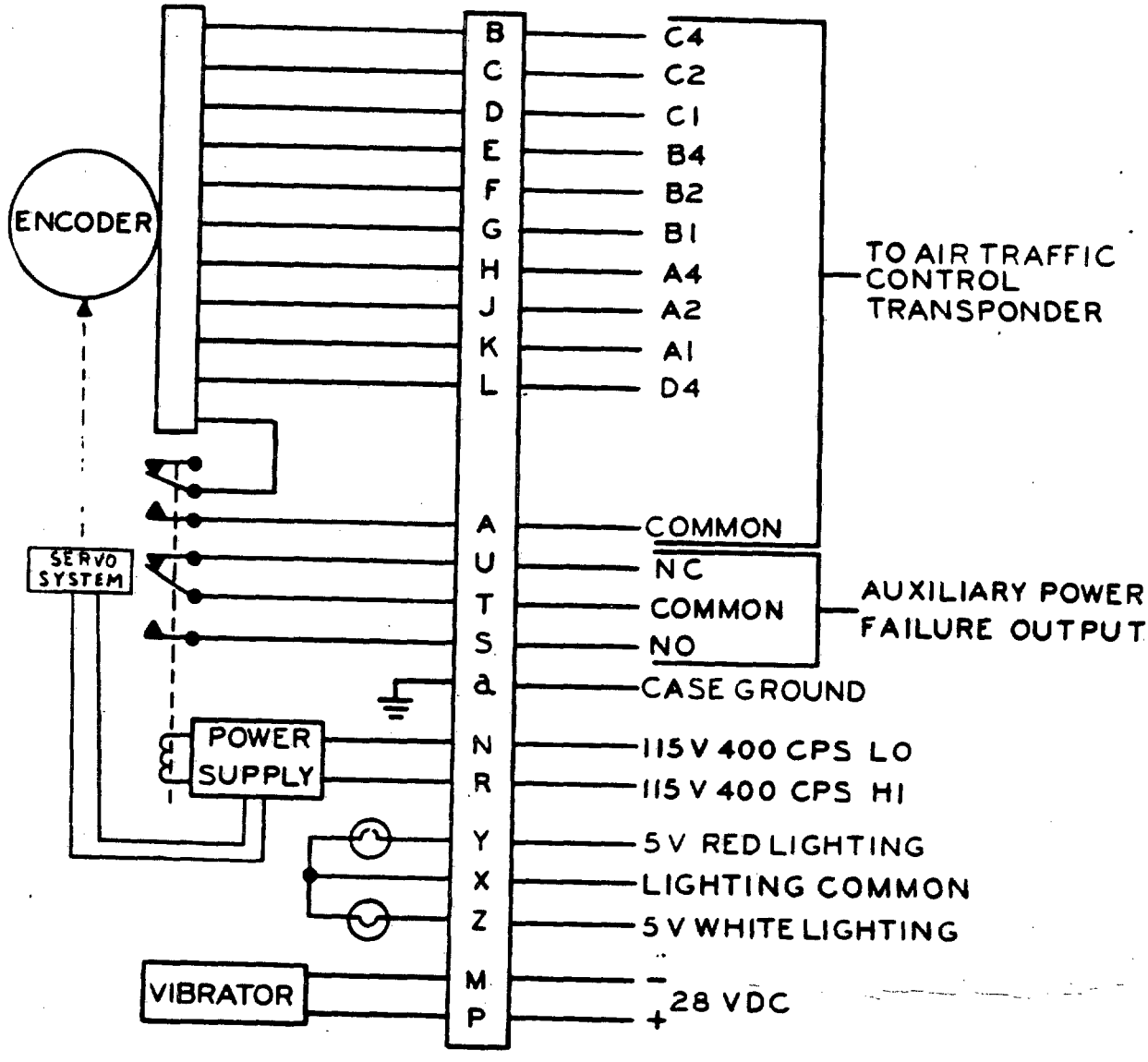


FIG. 3

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3.5.8 Relay Contact Ratings - The rating of the relay contact, switching the digitizer common lead, shall be at least .05 amps. The rating of the other contacts shall be at least 115V AC at .25 amps or 28V DC at 1 amp into either resistive or relay coil load.

3.5.9 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 seconds during normal electric operation. Operation shall return to normal with no resulting damage to the equipment.

3.5.10 Internal Vibrator - An internal vibrator shall be incorporated into the altimeter which shall effectively decrease the friction in the mechanism. The vibrator shall be self-starting upon being energized, and shall start without being tapped or other external vibration assistance. The vibrator unit shall be capable of starting, running and or producing vibration sufficient to meet the friction and reliability requirements specified herein for a period of 1,000 hours without failure. The design shall be such that extended use of the vibrator will not result in deposits of foreign material nor produce deleterious fumes that may result in adverse effects on altimeter performance.

3.5.10.1 Power - The internal vibrator circuit shall be designed to operate normally with the power requirements of MIL-STD-704, category B, for 28V d-c power. The input current shall not exceed 90 milliampers.

3.6 Dial - The dial shall be marked as shown in Figure 2. Unless otherwise specified, all visible portions of thedial shall be lusterless black Color No. 37038 of FED-STD-595, and all markings on the dial shall be lusterless white Color No. 37875 of FED-STD-595. The dimensions of the dial markings shall be as shown on Figure 2. The dial shall be securely fastened to the mechanism frame.

3.6.1 Visibility of Dial - The pointer, numerals, at least 1/32 inch of the shortest graduations, and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 20 degrees with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.6.2 Markings - All markings shall be durable so as to withstand the usage encountered in service. The style and proportions of numeral and letters placed on the dial shall conform to MS33558.

3.6.3 Numerals - Each numeral shall distinctly indicate the graduation to which it applies. If practicable, each numeral shall be so placed that the center mass of the numeral is on the radial line joining the appropriate graduation and the center of the dial. When several numerals are used in one group, the space between the numerals shall be approximately 1/64 inch.

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3.6.4 Finish and Dimensions - The following markings shall be finished in lusterless white, color No. 37875 of FED-STD-595. The dimensions shall be as specified.

Markings	Height or Length Inch ± 0.015	Width of Line or Graduation Inch ± 0.005
Numerals 0, 1, 2, 3, 4, 5,6,7,8,9, on counters and 0,1,2,3,4,5,6,9 on dial	0.281	0.035
Numerals 7, 8 on dial	0.187	0.023
100-foot graduation	0.172	0.030
50-foot graduation	0.109	0.015
Baro counter numerals	0.156	0.025
ALT lettering	0.188	0.025
IN. Hg Lettering	0.046	0.008
100 FEET lettering	0.046	0.008
1000 FEET lettering	0.046	0.008
Shaded portion of pointer	-----	-----
Border on altitude counter window	-----	0.035

The following markings shall be finished in lusterless black, color No. 37038 of FED-STD-595. The dimensions shall be as specified.

Markings	Height or Length Inch ± 0.015	Width of Line or Graduation Inch ± 0.005
38,000 FT lettering	0.062	0.011
CODE OFF Lettering	0.093	0.015
Unshaded portion of pointer, dial, counters	-----	-----
Numerals 0 through 9 on 100 ft. drum	0.156	0.025
AAU-21/A lettering	0.062	0.011

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The background of the CODE OFF flag shall be finished in Schweitzer Brothers Glo-Craft Red Fluorescent No. 66, formula No. 116 1548, or equivalent. The background of the 100 ft drum shall be finished in lusterless grey, color No. 36440 of FED-STD-595.

3.7 Integral Lighting - The altimeter-encoder shall be provided with integral lighting in accordance with Specifications MIL-L-25467 and MIL-L-27160 to illuminate the display. The lighting circuit shall operate from 5V AC or DC and shall be connected as per Figure 3.

3.7.1 Coverglass and lighting wedge - The coverglass and/or lighting wedges shall conform to DD-G-451 and shall not interfere with the readability of the instrument. It shall be properly sealed and shall be replaceable by removing the bezel. All reflecting glass surfaces shall be provided with a reflection-reducing coating that meets the requirements of MIL-C-675, in addition to withstanding the environmental conditions specified herein, except that the following reflectance tolerances shall apply:

Angle of Incidence	Wavelength-Millimicrons	Percent Reflectance
0°	450-675	0.6 absolute
0°	425-700	0.5 average
30°	450-625	1.0 absolute
30°	425-700	0.5 average

3.7.2 Distance from Dial to Coverglass - The distance from dial to the innermost glass surface shall not exceed 0.250 inch.

3.8 Radio Noise Interference - The altimeter-encoder shall be provided with such necessary noise suppression features that it will meet the limits specified for conducted and radiated radio noise in MIL-I-6181. In particular, these provisions shall include adequate filters and enclosed case construction which will minimize leakage through joints, seams, and mating surfaces. The volume and weight of the filtering components required shall be minimized by the application of proper electrical and mechanical design and construction.

3.9 Performance - The instrument shall be designed to meet the performance specified herein under the following conditions:

a. Temperatures - Continuous operation at temperatures ranging from -54° to +71°C and storage at temperatures ranging from -62° to +71°C.
b. Humidity - Relative humidity up to 100 percent.
c. Altitude - Pressure altitudes ranging from -1,000 to +38,000 feet.
d. Salt Spray - Exposure to a moist salt laden atmosphere.
e. Vibration - Vibration as per para. 4.7.2.7
f. Fungus - Fungus growth as encountered in tropical climates.
g. Sand and Dust - Airborne sand and dust particles as encountered in desert operations.

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h. Shock - Shock as per para. 4.7.2.9.
i. Pressurization - Compartment pressurization up to an absolute pressure of 60 in.Hg.
j. Acceleration - Acceleration forces of 20g for a period of 30 seconds.

3.10 Reliability Program - The contractor shall establish a reliability assurance program in accordance with MIL-STD-785.

3.10.1 Operational Stability - The altimeter-encoder shall operate with satisfactory performance, continuously or intermittently, for a period of at least 1,000 hours without the necessity for readjustment of any controls.

3.10.2 Reliability in Mean Time Between Failure (MTBF) - The altimeter-encoder shall have 1,000 hours MTBF of mean (operating) time between failures when tested and accepted as outlined under the requirements of para. 4.4.3.

3.10.3 Longevity - The altimeter-encoder shall have a total operating life of 3,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.11 Identification Markings shall be in accordance with MIL-N-18307. A nameplate shall be securely attached to the exterior of the case.

3.12 Weight - The weight of the complete assembled altimeter-encoder, without connecting fittings and mounting screws, shall not exceed 3.75 pounds.

3.13 Dimensions and Tolerances - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the altimeter-encoder, they shall be held or limited accordingly.

3.14 Interchangeability - All parts having the same maufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000/1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and service conform to prescribed requirements.

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4.2 Classification of Inspection - Inspection of the altimeter-encoder indicators shall be classified as follows:

(a) Qualification inspection: Qualification inspection consists of examinations and tests performed on sample indicators submitted for approval as a qualified product.
(b) Qualification conformance inspection: Quality conformance inspection consists of examinations and tests performed on indicators manufactured and submitted for acceptance under contract.

4.2.1 Qualification Inspection - The qualification inspection of the altimeter-encoder shall consist of all the examinations and tests of this specification performed in the order specified under the paragraphs 4.5, 4.6, 4.7 and 4.8, and shall include the reliability test for the Reliability Qualification Phase.

4.2.1.1 Qualification Inspection Sample - Qualification inspection samples shall consist of three indicators manufactured in accordance with this specification. The indicators submitted for qualification inspection shall have been previously subjected only to the inspections of 4.5 and 4.6. Samples and data shall be forwarded at Contractor's expense to the laboratory designated in the Letter of Authorization.

4.2.1.1.1 Qualification Inspection Sample Identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for Qualification Inspection
Altimeter-Encoder Type AAU-21/A
Submitted by (Manufacturer's name, date)
for Qualification Inspection in accordance
with Specification MIL-A-81403(AS) under
authorization (reference letter authorizing tests).

4.2.2 Quality Conformance Inspection - The quality conformance inspection shall consist of the Individual Inspection, the Sampling Plans, and the Reliability Assurance Tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing all the inspections. Sampling plan B inspections shall be conducted as directed by the procuring activity. Quality conformance inspection shall be under the supervision of the Government quality control representative. The contractor shall furnish test reports showing quantitative results for all tests required by this specification (including detailed results of the examination of product test with respect to meeting each applicable specification requirement) signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

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4.3 Individual Inspection - Each altimeter-encoder submitted for acceptance shall be subjected to the individual tests to determine compliance with the requirements of material, workmanship and operational adequacy. As a minimum, each instrument accepted shall have passed the following inspections:

a. Examination of product
b. Barometric setting scale
c. Backlash
d. Case leakage
e. Encoder Transition
f. Friction
g. Position error
h. Altitude display scale error
i. Scale error at -35°C
j. Scale error at +45°C
k. Lighting (individual)

4.4 Sampling Plans - The sampling plans shall consist of Sampling Plan A and Sampling Plan B and Reliability Sampling phase test of MIL-STD-781. The inspection samples which have been subjected to Sampling Plan A inspection shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual inspections. Inspection samples which have been subjected to the Sampling Plan B inspection shall not be delivered on contract as usable production equipment.

4.4.1 Sampling Plan A Sample Selection - Sampling Plan A samples shall be selected at random in accordance with the following schedule:

Quantity Offered for acceptance	Quantity to be Selected for Inspection
First 15	1 (Zero when Sampling Plan B is invoked)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

When an instrument failure occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of each fifteen from those still on hand, and proceed with sampling as indicated above when the next quantity is offered for acceptance. One additional sample, selected at random, shall be tested from each monthly lot for shelf life as per para 4.5.11. If the lot size is less than 25 instruments, the quantity produced in two months shall be considered as one lot. Concurrent delivery shall be permitted during the storage period.

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4.4.1.1 Sampling Plan A Inspection - Each sample selected for Sampling Plan A inspection shall be subjected to the following tests:

a. Dielectric strength
b. Individual tests
c. Scale error at -54°C
d. Scale error at +71°C
e. Vibration error
f. Power variation
g. Lighting Tests (Sampling Plan A)
h. Response Test
i. Encoder Compatibility
j. Encoder Re-Zero

4.4.2 Sampling Plan B Instructions - Three samples shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. Unless otherwise directed, these samples shall be forwarded at the contractor's expense to a Government Laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

Altimeter-Encoder AAU-21/A
Submitted by (Manufacturer's name, date) for
production acceptance Sampling Plan B inspection
of MIL-A-81403 with Contract No. _____
Manufacturer's part number _____

4.4.2.1 Sampling Plan B Inspection - Each sample selected for Sampling Plan B testing shall be subjected to the following tests:

a. Sampling Plan A tests
b. Encoder test
c. Low temperature exposure
d. High temperature exposure
e. Overpressure test
f. Underpressure test
g. Pressurization Test
h. Vibration failure
i. Acceleration
j. Impact Shock
k. Magnetic property
l. Lighting (Sampling Plan B)
m. Radio noise interference
n. Humidity
o. Fungus
p. Salt Spray
q. Sand and Dust
r. Internal Examination

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4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781 for the Qualification Phase and Sampling Phase Tests. Altimeter-encoders selected for reliability assurance tests shall first have passed the Individual tests.

4.4.3.1 Reliability Qualification Phase - Prior to the acceptance of the altimeter-encoders under the contract or order, between 3 and 6 instruments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests" using test Level E except as specified herein. Criteria for Test Plan III shall be used. The instruments shall demonstrate a specified MTBF of 1,000 hours. Following completion of cycling the altimeter-encoder may be reset by means of the barometric setting adjustment mechanism. The amount of reset shall be limited to 100 feet and an equivalent displacement shall be accomplished on the encoder zero.

4.4.3.1.1 Duty Cycle of Test Level E shall have a 4.5 hour heated period in-lieu of the 2 hours stated in MIL-STD-781. Each operating hour shall consist of:

a. Exercise - 4 exercise cycles, each cycle consisting of a change in altitude from zero feet to 25,000 feet and return to zero feet within 14 minutes; all power shall be OFF 3 minutes each hour; one minute warm-up time is permitted.
b. Lighting - 57 minutes ON (red and white lighting to be ON separately and alternated).
c. Vibration - 10 minutes vibration of ±0.5g at any non-resonanct frequency between 20 and 60 cps during the exercise period.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the altimeter-encoder shall be tested as outlined in MIL-STD-781 (as modified herein), under the section entitled "Sampling Phase of Production Reliability Tests." For the Sampling Phase, Test Level E (as modified herein) shall be used.

4.4.3.2.1 Lot Size for Sampling Phase - Between 3 and 6 instruments, selected at random, shall be tested from each monthly lot for the Accept-Reject Criterial of Test Plan IV. If the lot size is less than 25 instruments, the quantity produced in two months shall be considered as one lot. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command Attn: Avionics Division. At any time that the test results show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc.

4.4.3.3 Longevity Tests - When specified in the contract, Longevity tests shall be performed in accordance with Test Plan XXVIII of MIL-STD-781 and the duty cycle herein except that 3000 hours operating time shall

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be performed on two samples. Altimeter-encoders used on the Reliability Assurance Tests shall be used until each has accumulated the required 3000 hours.

4.4.3.3.1 Performance Characteristics to be Measured During Longevity Tests and Reliability Qualifications - At the end of each 250 operating hours the altimeter-encoders shall be subjected to the Encoder Transition, Friction and Scale Error tests, of paragraph 4.5.4, 4.5.5 and 4.5.7, at the asterisked test points of tables II, III and IV. The internal vibrator shall be checked to ascertain proper starting characteristics during the tests of para 4.5.5. The instruments shall be observed each working day for any obvious failures such as rough pointer movement, encoder response, or failure of the integral lighting. Upon completion of test, the instruments shall again be tested for Encoder Transition, Friction and Altitude Display Scale Error as per paragraphs 4.5.4, 4.5.5 and 4.5.7. Transient out-of-tolerance conditions will be permitted to the extent that a maximum of 20 percent of the test points may be out of tolerance by not more than 20 percent if the out-of-tolerance conditions are not repeated during the next regularly scheduled test period.

4.4.3.4 Failure Criteria - Whenever the measured Performance Characteristics of paragraphs 4.5.4, 4.5.5 and 4.5.7 fall below the acceptable requirements, at least one failure has occured. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one part caused the other parts to fail.

4.4.3.5 Preventive Maintenance - No preventive maintenance shall be performed on the altimeter-encoder during the 1000 hour MTBF test. The failures occuring during the longevity test shall be analyzed in accordance with MIL-STD-781. In addition, the analysis of failures, especially for what appears to be pattern type failures, should indicate whether preventive maintenance would have prevented the failure and extended the service life.

4.4.3.6 Disposition of Samples - Any sample tested may be delivered on contract provided the sample is in "good as new" condition or has been refurbished and meets the Individual Inspection Tests, and is representative of production units currently being accepted.

4.4.4 Test Conditions

4.4.4.1 Standard Conditions - Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	:	Room ambient, 25°C ± 5°C
Pressure	:	Normal atmospheric (approx. 29.92 inches Hg.)
Humidity	:	Room ambient 40 to 90% relative humidity
Power	:	115 V AC ± 1 volt, 400Hz. 5 ± 0.1 V AC or DC (for lighting) 28 V DC ± 1 volt (for integral vibration)

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4.4.4.2 Test Readings - Unless otherwise specified, the integral vibrator and the servo system shall be energized during all tests. In addition, the status of the relay contacts shall be monitored to determine satisfactory performance of the failure warning relay during each operational test .

4.4.4.3 Attitude - Unless otherwise specified, the altimeter-encoder shall be tested in its normal operating position.

4.4.4.4 Encoder Load - Unless otherwise specified the encoder shall be loaded to simulate operating conditions during all tests specified herein. The load shall be a purely resistive load or a simulated transponder circuit. In either case the current through the encoder contacts shall be 2.5 ma.

4.4.4.5 Integral Lighting - The altimeter-encoder lighting system shall be energized and shall operate satisfactorily during all tests except humidity, fungus resistance, salt spray, sand and dust, and as otherwise specified herein. The lighting shall operate satisfactorily after completion of each test unless otherwise specified herein.

4.4.4.6 Rate of Pressure Change - Unless otherwise specified, the rate of change in pressure during all tests shall be such as to produce the following changes in height indications:

Decreasing pressure	6,000 ft/min
Increasing pressure	6,000 ft/min

This rate shall be progressively reduced as the checkpoints are approached to avoid passing the checkpoint. The movement of the pointer during decreasing and increasing pressures shall be smooth and free from irregular motion when friction is removed as specified in 4.4.4.2.

4.5 Individual Tests

4.5.1 Examination of Product - Each altimeter-encoder shall be examined externally to determine conformance with the applicable drawings and with each of the requirements of this specification capable of being inspected in this manner.

4.5.2 Barometric Setting Scale - The altimeter-encoder shall be subjected to atmospheric pressure. The pressure scale shall be set successively to the values specified in Table I.

After the series of readings has been recorded, the difference between the reading at the setting of 29.92 and each other setting, shall be found. These differences shall be recorded and compared with the values in the "correct differences" column of Table I. The recorded values shall not differ from the correct differences by more than 25 feet at any setting. Atmospheric pressure shall be recorded before and after this test and proper allowance shall be made for any variation.

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The encoder output and the failure warning flag shall be observed during this test. The OFF flag shall not be effected. The encoder signal may fluctuate by 100 feet while the setting knob is rotated. If this occurs the pressure altitude shall be changed by a maximum of 50 feet and the test shall be repeated. The encoder signal shall not fluctuate/change while the setting knob is rotated.

TABLE I

BAROMETRIC SETTING SCALE

Pressure Scale Setting	Correct Difference
28.1	-1,727
28.5	-1,340
* 29.0	-863
* 29.5	-392
29.92	0
30.5	531
* 30.9	893
31.0	983

4.5.2.1 Backlash - This test may be conducted in conjunction with the test specified in 4.5.2 by approaching three separate points from both directions. The pressure scale knob shall be turned in a clockwise direction to a specific setting and the height indication noted. Turning of the knob in a cw direction shall be continued for approximately 200 feet. The knob shall be slowly turned in a ccw direction until the previous setting has been reached. The pointers shall indicate the previous height within 20 feet and shall remain within the barometric setting tolerances using the test points specified in Table I.

4.5.3 Case Leakage - A vacuum sufficient to produce a reading of approximately 30,000 feet shall be slowly applied to the static pressure connection of the altimeter-encoder, at which point the connecting tubing shall be pinched off or otherwise completely sealed. The change in reading in 1 minute shall not exceed 100 feet. Altimeter-encoders shall be tested individually.

4.5.4 Encoder Transition - The encoder shall be connected to a suitable readout device to monitor the encoder transition points. During the test the encoder shall be monitored to determine that the requirements of paragraph 3.5.6.1 and paragraph 3.5.6.2 (room temperature only) are met. The barometric setting shall be set at 29.92 in. Hg and a pressure altitude of -1,000 feet shall be applied to the altimeter-encoder. The altitude shall then be increased at a rate not to exceed that specified in paragraph 4.4.4.6 except that through each of the encoder transition test points in column A of table II the rate shall not exceed 500 ft/min. The altimeter-encoder

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display shall be carefully read as the appropriate transition test point of the code is indicated. The instrument display shall not differ by more than 40 feet from the recorded transition point indicated altitude value. This test shall be repeated except that the test shall start from 38,000 feet and the altitude shall be decreased. The transition points for this test are listed in column B of Table II. Throughout the range of operation no disrupted signals in encoder output shall be noted and every altitude increment shall be present.

TABLE II

	Transition Altitude (±40 ft. Pointer Display)	Encoder Output A From	A To	B From	B To
	-950	-1,000	-900	-900	-1,000
*	50	0	100	100	0
*	2,050	2,000	2,100	2,100	2,000
	4,050	4,000	4,100	4,100	4,000
*	6,050	6,000	6,100	6,100	6,000
	8,050	8,000	8,100	8,100	8,000
*	10,050	10,000	10,100	10,100	10,000
	15,050	15,000	15,100	15,100	15,000
*	20,050	20,000	20,100	20,100	20,000
	25,050	25,000	25,100	25,100	25,000
	30,050	30,000	30,100	30,100	30,000
*	35,050	35,000	35,100	35,100	35,000
	38,050	38,000	38,100	------	------

4.5.5 Friction Stop and Jump - The altimeter-encoder shall be tested for friction in both ascending and descending directions at the test points specified in Table III. With the internal vibrator in operation, the servo power on and the encoder energized, the altimeter-encoder shall be subjected to a constant rate of decreasing and increasing pressure equivalent to the rate indicated at the test points in Table III. The change in reading of the pointer at the test points, due to its stopping and jumping as the counter is turning, shall be recorded as friction, and shall not exceed the tolerance specified in Table III Column C. This test shall then be repeated with the internal vibrator inoperative. The stop and jump friction shall not exceed the tolerances specified in Table III, Column D.

4.5.5.1 Static Friction - At the completion of the test above, the altimeter-encoder, with the vibrator not operating, the servo power off, and the encoder not energized, shall be subjected to a constant rate of decreasing pressure not exceeding 3,000 fpm until the pressure reaches the equivalent of ± 100 feet of the test points listed in Table III under Static Friction

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Error, at which time the rate of pressure change shall be gradually shut off as the pressure equivalent of the desired test point is reached, then held constant while two readings are taken, the first without the vibrator operating and the second with the vibrator operating. The difference between the two readings shall be recorded as static friction and shall not exceed the tolerances listed in Table III.

TABLE III

STOP AND JUMP FRICTION ERROR

COLUMN "A" Test Point Altitude (Feet)	COLUMN "B" Rate of Ascent or Descent (Feet/Min)	COLUMN "C" Tolerance Vibrator Operative (Feet)	COLUMN "D" Tolerance Vibrator Inoperative (Ft.)
0	500	25	300
1,000	500	25	120
3,000	500	25	120
5,000	500	25	120
10,000	500	25	350
20,000	3000	25	350
30,000	3000	25	350
35,000	3000	50	600

STATIC FRICTION ERROR - VIBRATOR INOPERATIVE

Test Point Altitude (Feet)	Tolerance (Feet)
500	70
1,500	70
2,500	70
3,500	70
7,500	80
12,500	90
17,500	110
22,500	120
32,500	180
35,500	300

4.5.6 Position Error - The altimeter-encoder position error readings shall be taken while the instrument is vented to atmosphere and in each of the following positions:

a. Normal operating position
b. Rotated about the longitudinal axis 90°
c. Rotated about the longitudinal axis 180°
d. Rotated about the longitudinal axis 270°
e. Dial face up

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The readings in positions b, c, d, e, or any other position, shall not differ from the reading in the normal operating position by more than 20 feet. This test shall be performed in the power OFF and power ON modes.

4.5.7 Altitude Display Scale Error - For a period of not less than 12 hours prior to this test, altimeter-encoder shall not have been subjected to any testing involving operation at other than atmosphere pressure. The instrument shall be tested for scale errors as follows: The barometric setting shall be set at 29.92 in. Hg and shall remain at this setting during all scale error tests. The instrument shall be subjected throughout the range to successively decreasing pressure and then increasing pressure as specified in Table IV.

The instrument shall remain at the pressure corresponding to each test point for at least one but not more than five minutes before a test reading is taken. The scale error shall not exceed the tolerances specified in Table IV. At each test point after the reading has been recorded, the servo power shall be turned off. When changing from power on to off, the pointer shall not change its indication more than 10 feet. The power shall be turned on before proceeding to the next altitude point. This test may be conducted with the instrument in an altitude chamber. There shall be no encoded output when the servo power is off. A switch controlling the altimeter servo power shall be located in the external circuit for this test. The movement of the counter, drum, and pointer shall be free from backlash and any irregular motion, when pressure is varied uniformly.

4.5.7.1 Hysteresis and After Effect - Tests of this paragraph may be combined with tests of paragraph 4.5.7. Not more than 15 minutes after the altimeter-encoder has been first subjected to the pressure corresponding to 35,000 feet in the test specified in 4.5.7, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 fpm until the pressure corresponding to the first test point given in Table V is reached. The instrument shall remain at this pressure for at least 1 but not more than 15 minutes before the test reading is taken. The pressure shall again be increased at the same rate until the pressure corresponding to the second test point in Table V is reached. The instrument shall remain at this pressure for at least 1 but not more than 10 minutes before the test reading is taken. The pressure shall again be increased at the same rate until 0 feet is reached. The reading of the instrument at the three test points shall not differ from the reading of the instrument for the same test points in the tests specified in 4.5.7 by more than the tolerance specified in Table V. In addition, all test readings shall be within the tolerance of Table IV.

4.5.8 Scale Error at -35°C - The tests specified under paragraphs 4.5.3, 4.5.4 and 4.5.7, shall be repeated, except that during the tests and for a period of 3 hours without power before the test, the altimeter-encoder shall be at a temperature of -35°C. The tests shall be initiated within 2 minutes after power is applied. The failure warning flag and the encoder shall operate satisfactorily during the test. The test points and tolerances shall be as specified in Tables II and IV.

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NOTE: The 38,000 Ft. Test Point is for ascending altitude only.

TABLE IV

Scale Errors

Equivalent Pressure Units of Mercury at 0°C		Standard Altitude Feet	Scale Error Tolerance Feet		
				Room Temp.	
MM	Inches		-35°C	+45°C	-54°C +71°C
787.87	31.018	-1,000	±45	±30	
*760.00	29.921	0	45	30	±60
746.37	29.385	500	50	35	
732.93	28.856	1,000	50	35	
719.70	28.335	1,500	60	40	
*706.65	27.821	2,000	60	40	80
693.80	27.315	2,500	70	45	
681.14	26.817	3,000	70	45	
656.38	25.842	4,000	75	50	
*632.36	24.896	5,000	75	50	100
609.05	23.978	6,000	75	55	
564.51	22.224	8,000	80	60	
*522.65	20.577	10,000	85	65	130
483.34	19.029	12,000	90	70	
446.46	17.577	14,000	95	75	
428.90	16.886	15,000	100	80	160
411.90	16.216	16,000	100	85	
379.53	14.942	18,000	105	90	
*349.25	13.750	20,000	110	95	190
320.96	12.636	22,000	115	100	
282.03	11.103	25,000	120	105	210
225.69	8.885	30,000	135	115	230
*178.83	7.041	35,000	150	125	
154.85	6.097	38,000	160	140	

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4.5.9 Scale Error at +45°C - The tests specified in paragraph 4.5.8 shall be repeated except that the temperature shall be +45°C.

4.5.10 Lighting - Individual tests required in MIL-L-25467 and MIL-L-27160 shall be performed. Combining tests in these specifications is permitted.

4.5.11 Shelf Life - The altimeter-encoder shall be placed in storage for a period of not less than 30 days. During the 30 day period, the instrument shall remain as dormant as practical. The mechanism shall not be exercised other than that caused by normal atmospheric pressure variation. The temperature shall remain constant; however normal room temperature variations will be permitted. At the end of the 30 day period, the altimeter-encoder shall be subjected to and meet the Individual Test of paragraph 4.3.

TABLE V

HYSTERESIS

Test Point	Std. Alt.	Permissible Variation
1st.	25,000 ft.	50 ft.
2nd.	20,000 ft.	50 ft.
3rd.	0 ft.	30 ft.

4.6 Sampling Plan A Tests

4.6.1 Dielectric Strength - The altimeter-encoder shall be checked for insulation resistance by testing from the terminals indicated to ground at the potentials indicated for no more than 30 seconds.

NOTE: The maximum voltage shall be raised from zero by rheostat control, switching to the voltage from zero shall not be permitted.

UTS	together	500V
NR	together	500V
XYZ	together	500V

There shall be no breakdown of insulation or any other permanent damage to the indicator as a result of this test.

4.6.2 Individual Tests - The tests specified in paragraph 4.3 shall be performed.

4.6.3 Scale Error at -54°C - The tests specified under paragraph 4.5.4 and 4.5.7 shall be repeated except that during the test and for a period of 3 hours without power applied before the test, the altimeter-encoder shall be at a temperature of -54°C ± 2°C. The test shall be initiated within 2 minutes after power is applied. The test points and tolerances shall be as specified in Tables II and IV.

4.6.4 Scale Error at +71°C - The test specified in paragraph 4.6.3 shall be repeated except the temperature shall be +71°C ±2°C

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4.6.5 Vibration Error - The altimeter-encoder shall be mounted on the vibration stand in its normal operating position and secured by flange screws in the same position as on an aircraft panel. The instrument shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch at frequencies from 500 to 3,000 cpm so that any point on the instrument case will describe a circular path of the amplitude stated, inclined at an angle of 45° to the horizontal and dial planes. The natural frequency of the instrument shall not occur in this frequency range. While the instrument is being vibrated, the maximum amplitude of the pointer oscillation shall not exceed 20 feet and the pointer variation from the normal position shall not exceed 20 feet. With the altitude set at 30 feet off of the code transition point between 20,200 and 20,300 feet, and while under vibration, the encoder shall not have intermittent output. No shift in the barometric setting shall occur with respect to the setting established prior to vibration. The integral lighting of the instrument shall be operated during this test and shall not flicker. While the instrument is being vibrated, the drum shall not exhibit oscillatory motion greater than 20 feet.

4.6.6 Power Variation - The altimeter-encoder, when connected electrically, shall be tested for accuracy at the asterisked test points specified in Table II at the following voltage-frequency combinations: 108V,320Hz; 108V,480Hz; 115V, 400 Hz; 121V, 320 Hz; 121V, 480 Hz. The change in the encoder transition from the normal power combination (115V, 400Hz) shall not exceed 20 feet. Operation at any frequency outside of MIL-STD-704 is to have a duty cycle of 6 minutes out of 1 hour.

4.6.7 Lighting Tests - Lighting tests specified as Sampling A tests in MIL-L-25467 and MIL-L-27160 shall be conducted. Those tests which are common may be conducted once.

4.6.8 Response Test - The altimeter-encoder shall be subjected to a rate of change of pressure equivalent to 20,000 ft/min. The encoder output shall not lag the dial display by more than 100 feet. No damage to the internal mechanism shall occur due to this test and no zero or calibration shift shall result.

4.6.9 Encoder Compatibility - The altimeter-encoder shall be tested to determine that the requirements of paragraph 3.5.6.1 and paragraph 3.5.6.2 are met.

4.6.10 Encoder Re-Zero - The altimeter-encoder shall be tested to determine that the requirements of paragraph 3.5.5 are met.

4.7 SAMPLING PLAN B

4.7.1 Sampling Plan A Tests - The tests specified in paragraph 4.4.1.1 shall be performed.

4.7.2 Environmental Tests - Unless otherwise specified, the altimeter-encoder shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-E-5272. Except after the tests specified in paragraph 4.7.2.11 and 4.7.2.12, the instrument shall meet

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the tests of 4.5.3, 4.5.4, 4.5.5, 4.5.7 and 4.5.10 at the asterisked points upon completion of each environmental test. When external connections are required to simulate installed conditions, they shall consist of standard AN or MS fittings and at least 10 feet of 1/4-inch copper tubing coiled and so arranged as to allow drainage of condensed vapors. They shall be attached to the static pressure port of the instrument.

4.7.2.1 Encoder Test - The test specified in paragraph 4.5.4 shall be repeated except that the test points of Table VI shall apply and the test performed for ascending altitude only.

4.7.2.2 Low Temperature Exposure - The low temperature exposure test shall be conducted in accordance with Procedure II.

4.7.2.3 High Temperature Exposure - The high temperature exposure test shall be conducted in accordance with Procedure I, except that the temperature shall be 71° ± 2°C.

4.7.2.4 Overpressure Test - With the barometric setting set at 29.92, a pressure of 35 in. Hg absolute shall be applied to the static fitting of the altimeter-encoder. No damage to the internal mechanism shall occur due to this pressure and no zero or calibration shift shall result.

4.7.2.5 Underpressure Test - The altimeter-encoder mechanism shall be capable of withstanding exposure to an altitude of 50,000 feet. No damage to the internal mechanism shall occur due to this pressure and no zero or calibration shift shall result.

4.7.2.6 Pressurization Test - The altimeter-encoder shall be placed in a pressure chamber with the pressure port connected to outside ambient pressure (29.92 inches of mercury approx.) during this test. The chamber pressure shall be raised to 60 in. Hg absolute and maintained at approximately 25°C for a period of 4 hours. During the 4 hour period, no deformation or other damage to the instrument shall occur. At the conclusion of this test, the instrument shall be closely examined and any evidence of deformation, damage, or unsatisfactory operation shall be cause for rejection.

4.7.2.7 Vibration Failure - The vibration test shall be conducted in accordance with Procedure V.

4.7.2.8 Acceleration - The acceleration test shall be conducted in accordance with Procedure I, except that the altimeter-encoder shall be mounted in its normal operating position with the rear of the case toward the center of rotation, and shall be subjected to an acceleration of 20 g for a period of 30 seconds. This test shall be repeated with the instrument mounted successively in the following two positions: (1) rotated 90°C in azimuth with respect to the original position, and (2) dial horizontal.

4.7.2.9 Impact Shock - The shock test shall be conducted in accordance with Procedure V, operational shock only.

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TABLE VI

TRANSITION ALTITUDE ± 40 FT. POINTER DISPLAY	CHANGING BIT	ENCODER READOUT FROM	TO
-950	C1 ON	-1,000	-900
-850	C2 OFF	-900	-800
-750	B4 ON	-800	-700
-650	C2 ON	-700	-600
-550	C1 OFF	-600	-500
-450	C4 ON	-500	-400
-350	C2 OFF	-400	-300
-250	B2 ON	-300	-200
-150	C2 ON	-200	-100
-50	C4 OFF	-100	0
50	C1 ON	0	100
150	C2 OFF	100	200
250	B4 OFF	200	300
350	C2 ON	300	400
450	C1 OFF	400	500
550	C4 ON	500	600
650	C2 OFF	600	700
750	B1 ON	700	800
850	C2 ON	800	900
950	C4 OFF	900	1000
1050	C1 ON	1000	1100
1150	C2 OFF	1100	1200
1250	B4 ON	1200	1300
1350	C2 ON	1300	1400
1450	C1 OFF	1400	1500
1550	C4 ON	1500	1600
1650	C2 OFF	1600	1700
1750	B2 OFF	1700	1800
1850	C2 ON	1800	1900
1950	C4 OFF	1900	2000
2050	C1 ON	2000	2100
2150	C2 OFF	2100	2200
2250	B4 OFF	2200	2300
2350	C2 ON	2300	2400
2450	C1 OFF	2400	2500
2550	C4 ON	2500	2600
2650	C2 OFF	2600	2700
2750	A4 ON	2700	2800
2850	C2 ON	2800	2900
2950	C4 OFF	2900	3000
3050	C1 ON	3000	3100
3150	C2 OFF	3100	3200
3250	B4 ON	3200	3300
3350	C2 ON	3300	3400
3450	C1 OFF	3400	3500
3550	C4 ON	3500	3600

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TABLE VI (Cont)

TRANSITION ALTITUDE ± 40 FT. POINTER DISPLAY	CHANGING BIT	ENCODER READOUT FROM	TO
3650	C2 OFF	3600	3700
3750	B2 ON	3700	3800
3850	C2 ON	3800	3900
3950	C4 OFF	3900	4000
4050	C1 ON	4000	4100
4150	C2 OFF	4100	4200
4250	B4 OFF	4200	4300
4350	C2 ON	4300	4400
4450	C1 OFF	4400	4500
4550	C4 ON	4500	4600
4650	C2 OFF	4600	4700
4750	B1 OFF	4700	4800
4850	C2 ON	4800	4900
4950	C4 OFF	4900	5000
5050	C1 ON	5000	5100
5150	C2 OFF	5100	5200
5250	B4 ON	5200	5300
5350	C2 ON	5300	5400
5450	C1 OFF	5400	5500
5550	C4 ON	5500	5600
5650	C2 OFF	5600	5700
5750	B2 OFF	5700	5800
5850	C2 ON	5800	5900
5950	C4 OFF	5900	6000
6050	C1 ON	6000	6100
6150	C2 OFF	6100	6200
6250	B4 OFF	6200	6300
6350	C2 ON	6300	6400
6450	C1 OFF	6400	6500
6550	C4 ON	6500	6600
6650	C2 OFF	6600	6700
6750	A2 ON	6700	6800
6850	C2 ON	6800	6900
6950	C4 OFF	6900	7000
10750	A4 OFF	10700	10800
14750	A1 ON	14700	14800
18750	A4 ON	18700	18800
22750	A2 OFF	22700	22800
26750	A4 OFF	26700	26800
30750	D4 ON	30700	30800
34750	A4 ON	34700	34800
38750	A2 ON	38700	38800

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4.7.2.10 Magnetic Property - The altimeter-encoder, not energized, shall be rotated in a vertical plane about a short-bar magnet compass with the nearest part of the altimeter-encoder 5 1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the instrument shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the instrument shall be rotated about on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. With the altimeter-encoder at any specified position, th compass deflection shall not exceed 3°. This test shall be repeated with the instrument energized with rated voltage.

4.7.2.11 Lighting - Lighting tests shall be conducted in accordance with sampling plan B tests of MIL-L-25467 and MIL-L-27160, as specified. Those tests which are common to both specifications may be conducted once.

4.7.2.12 Radio Noise Interference - The altime.er-encoder shall be subjected to a radio noise interference test in accordan'e with MIL-I-6181. During this test the instrument shall be pressure cycled :arc ghout a range that will exercise all the outputs. The RFI Tests shall be conducted under the conditions specified in 4.4.4.4.

4.7.2.13 Humidity - The humidity test shall be conducted in accordance with Procedure I. The altimeter-encoder shall be thoroughly examined within at least 48 hours after completion of the humidity exposure, and there shall be no evidence of corrosion and deterioration that would affect subsequent operation of the instrument.

4.7.2.14 Fungus - The fungus test shall be conducted in accordance with Procedure I. At the end of the test period, the altimeter-encoder shall be thoroughly examined to ascertain that no fungus growth has occurred.

4.7.2.15 Salt Spray - The salt spray test shall be conducted in accordance with Procedure I for a period of 50 hours. There shall be no corrosion or other damage as a result of this test that would affect subsequent operation of the altimeter-encoder.

4.7.2.16 Sand and Dust - The sand and dust test shall be conducted in accordance with Procedure I.

4.7.2.17 Internal Examination - The case of the altimeter-encoder shall be opened and the internal mechanism shall be examined to determine conformance with the applicable requirements of the specification. The lamp circuit of the altimeter-encoder shall also be examined for conformance to the requirements of MIL-L-25467. In addition, any damage or deterioration to the mechanism that resulted from any of the preceding tests, or any other deficiency which could in any manner prevent the instrument from meeting operational or maintenance requirements during service life shall be cause of rejection.

4.7.3 Rejection and Retest - When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After corrections have been made, all necessary tests shall be repeated.

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4.8 Maintainability Demonstration - Maintainability Demonstration shall be in accordance with MIL-STD-471 except as specified herein. Maintainability Demonstration shall be performed on a minumum of three selected repairs. The tasks selected to demonstrate maintainability shall be selected in advance of timing the tasks and shall be approved by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The altimeter-encoder shall be packaged in accordance with MIL-STD-794, Level A or C as specified in contract or order. For Level A packaging, the method of preservation shall be in accordance with Specification MIL-P-116 Method 1A-5 without preservation compound, using reusable metal containers.

5.2 Packing - The altimeter-encoder shall be packed in accordance with MIL-STD-794, Level A, B, or C, as specified in contract or order.

5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-794 with the additional precautionary marking appearing conspicuously on two opposite sides of each interior package and each shipping container: "FRAGILE - DELICATE INSTRUMENTS - HANDLE WITH CARE".

6. NOTES

6.1 Intended Use - The altimeter-encoder covered by this specification is intended for use in aircraft to indicate and transmit the pressure altitude of the aircraft above some reference point, assuming standard conditions of temperature and pressure.

6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether Sampling Plan B is to be omitted, or the name of the laboratory where samples shall be submitted.
(c) Whether Reliability Qualification Phase tests are required (See paragraphs 4.4.3 and 4.4.3.1).
(d) Reliability Sampling Phase Tests (See paragraphs 4.4.3 and 4.4.3.2).
(e) Longevity Tests (See paragraph 4.4.3.3).
(f) Maintainability Demonstration (See paragraph 4.8).
(g) Levels of packaging and packing desired. (See paragraphs 5.1 and 5.2).

6.3 Definitions

6.3.1 Good as new - "Good as new" shall mean the indicator has been operated less than 10 percent of specified MTBF operation.

6.3.2 Refurbished - "Refurbished" shall mean the indicator has been completely overhauled with all component parts meeting current parts standards;

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and the indicator shall have been subjected to, and met, all the requirements of a new indicator.

6.4 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity.

6.5 Precedence of Documents - When the requirement of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.
(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specification where applicable, shall be specifically approved in writing by the procuring activity.
(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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